

HAY RIVER, NT, Canada

WMO: 719350

Lat: 60.8394N

Lon: 115.7816W

Elev: 541

StdP: 14.41

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.2	-29.9	-42.2	0.5	-34.0	-38.1	0.6	-29.7	20.9	2.6	18.8	0.6	4.2	240	0.729

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	80.9	62.5	77.2	60.9	73.6	59.5	64.6	76.2	62.9	73.5	61.3	71.0	9.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	79.5	68.2	58.5	74.7	67.1	56.9	70.4	65.9	30.0	76.3	28.7	73.6	27.6	71.0	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.9	17.7	15.7	DB	-39.1	87.8	3.8	2.9	-41.8	89.9	-44.0	91.6	-46.2	93.3	-48.9	95.4
			WB	-39.2	67.7	3.8	2.0	-41.9	69.1	-44.1	70.3	-46.3	71.4	-49.0	72.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.4	-6.5	-2.0	7.2	26.6	42.5	55.4	62.0	58.7	48.8	33.8	13.0	-0.7	(6)
	DBStd	26.83	14.52	13.36	16.10	13.60	10.32	8.01	6.08	6.45	7.77	8.07	11.62	14.28	(7)
	HDD50	8862	1751	1455	1328	703	279	37	1	7	115	505	1110	1571	(8)
	HDD65	13432	2216	1875	1792	1151	699	300	133	215	488	967	1560	2036	(9)
	CDD50	977	0	0	0	2	46	200	373	275	78	3	0	0	(10)
	CDD65	73	0	0	0	0	1	13	40	18	1	0	0	0	(11)
	CDH74	707	0	0	0	1	31	140	327	188	19	1	0	0	(12)
	CDH80	134	0	0	0	0	4	22	67	39	2	0	0	0	(13)
	WSAvg	7.2	6.5	6.6	7.0	7.7	8.0	7.0	6.8	7.2	8.0	7.6	7.6	6.9	(14)
(15) Precipitation	PrecAvg	14.7	0.9	0.7	0.7	0.7	0.9	1.4	1.9	1.7	1.6	1.5	1.4	0.9	(15)
	PrecMax	20.7	2.0	1.7	2.2	3.0	2.4	5.1	4.3	3.9	4.9	4.4	4.5	2.5	(16)
	PrecMin	9.4	0.1	0.1	0.0	0.0	0.0	0.2	0.2	0.1	0.2	0.2	0.4	0.2	(17)
	PrecStd	3.5	0.5	0.4	0.5	0.7	0.6	1.1	1.1	0.9	0.9	1.0	0.9	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.5	39.9	49.6	65.8	78.0	83.3	85.9	85.0	76.8	61.1	43.9	35.9	(19)
		MCWB	32.6	34.1	39.6	49.1	56.1	61.9	64.4	64.7	60.1	50.0	37.1	30.9	(20)
	2%	DB	28.2	30.3	42.6	57.0	71.7	78.1	81.4	79.1	70.1	53.2	35.1	30.0	(21)
		MCWB	25.5	27.2	35.4	43.8	53.4	59.9	63.3	62.5	56.4	44.8	32.3	27.5	(22)
	5%	DB	20.4	23.1	37.4	50.7	65.4	73.7	77.6	74.7	65.5	48.6	31.7	24.0	(23)
		MCWB	19.0	21.1	32.0	40.7	50.1	57.9	62.2	61.0	54.2	42.6	30.2	22.6	(24)
	10%	DB	13.5	16.8	32.0	45.1	59.6	69.5	73.8	70.6	61.1	44.9	28.3	19.5	(25)
		MCWB	12.5	15.4	28.3	37.5	47.4	56.5	60.9	59.4	52.6	40.7	26.9	18.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	34.4	39.8	49.7	57.7	64.6	67.3	66.9	61.5	51.3	37.4	31.9	(27)
		MCDB	36.5	39.8	48.8	65.7	75.2	77.3	80.5	80.8	74.0	59.6	43.3	35.5	(28)
	2%	WB	25.3	27.2	35.6	44.5	54.0	61.7	65.1	64.1	58.0	46.1	32.5	27.5	(29)
		MCDB	28.0	30.4	41.8	55.7	69.4	73.7	76.8	75.3	67.2	51.6	34.3	30.0	(30)
	5%	WB	19.0	21.1	32.1	41.0	50.8	59.6	63.5	62.2	55.5	43.1	30.1	22.8	(31)
		MCDB	20.2	22.9	37.1	49.9	64.6	70.8	74.3	71.9	63.4	47.9	31.9	24.1	(32)
	10%	WB	12.5	15.5	28.2	37.8	47.8	57.5	62.1	60.4	53.4	40.6	26.9	18.4	(33)
		MCDB	13.4	16.8	32.1	44.8	58.8	67.6	71.9	69.0	60.3	44.8	28.3	19.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	17.2	19.1	18.6	18.2	17.9	16.6	16.6	15.6	11.3	11.8	13.5	(35)
		MCDBR	19.3	23.8	23.2	25.5	28.4	24.6	23.2	24.1	23.2	18.7	13.0	16.5	(36)
	5% WB	MCWBR	17.8	21.4	17.4	15.7	14.8	11.8	10.8	11.8	13.2	12.2	11.2	14.9	(37)
		MCWBR	19.4	23.3	22.2	24.3	27.4	21.4	19.6	21.3	20.5	16.9	12.7	16.4	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.210	0.241	0.267	0.321	0.353	0.356	0.378	0.351	0.313	0.274	0.209	0.160	(40)
		taud	2.101	2.277	2.294	2.342	2.407	2.423	2.396	2.499	2.552	2.504	2.160	1.995	(41)
	Ebn at Noon	Ebn at Noon	178	247	276	278	276	277	266	266	260	235	178	152	(42)
		Edh at Noon	12	20	27	32	33	33	33	28	22	16	11	8	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	118	392	956	1542	1848	1941	1757	1369	862	373	123	61	(44)
	RadStd	RadStd	10	25	53	89	124	154	102	69	73	37	17	6	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.97	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	+2.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page